

AROC

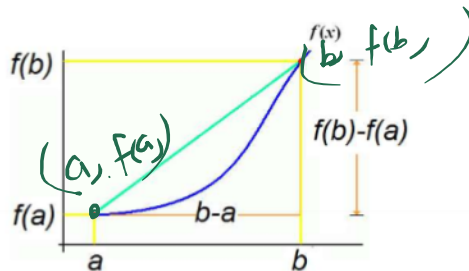
IROC

### 1.11 Average Rate of Change & Instantaneous Rate of Change

The average rate of change of  $f(x)$  on the interval  $[a, b]$  is defined as the slope of the secant drawn to the graph over the interval  $[a, b]$ .

$$\text{A.R.O.C} = m = \frac{f(b) - f(a)}{b - a}$$

Recall: A secant line is a straight line that joins two points on the function.



$$\text{AROC} = m = \frac{(x_2, y_2) - (x_1, y_1)}{x_2 - x_1} = \frac{\text{Rise}}{\text{Run}} = \frac{\Delta y}{\Delta x}$$

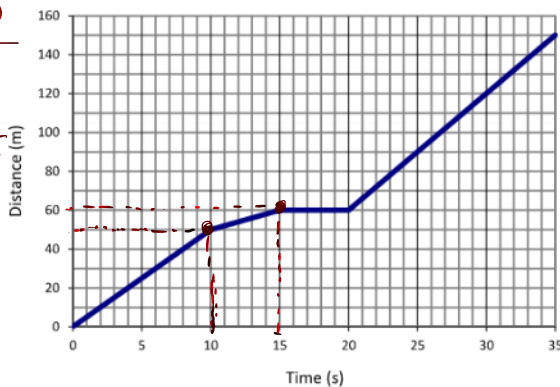
Example#1: Calculate the average rate of change

i. between 10s and 15s.

$$\begin{aligned} \text{AROC} = m &= \frac{f(15) - f(10)}{15 - 10} \\ &= \frac{60 - 50}{5} \frac{d}{t} \\ &= 2 \end{aligned}$$

ii. between 4s and 30s.

$$m = \frac{f(30) - f(4)}{30 - 4}$$



$$= 3.85$$

$\therefore$  The speed is 3.85 m/s.

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AROC  $\rightarrow$  slope

**Example #2** Determine the average rate of change of the secant line on the graph of  $g(x) = x^2 - 4x$  in interval  $[1, 5]$ .

$$\text{AROC} = \frac{f(b) - f(a)}{b - a}$$

$$= \frac{f(5) - f(1)}{5 - 1} \quad (2)$$

$$f(5) = 5^2 - 4(5)$$

$$f(5) = 5$$

$$f(1) = 1^2 - 4(1)$$

$$f(1) = -3$$

sub.  $f(5)$  &  $f(1)$  in (2)

$$\text{AROC} = \frac{5 - (-3)}{4}$$

$$= \frac{8}{4}$$

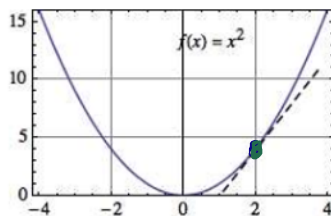
$$\text{AROC} = 2$$

### Instantaneous Rate of Change

The exact rate of change of a function  $y = f(x)$  at a specific value of the independent variable  $x = a$

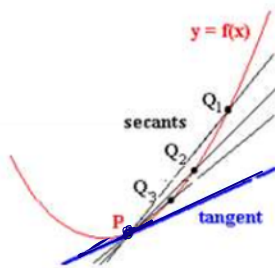
- Instantaneous rate of change** = slope of tangent

Recall: **Tangent Line** is a line which only touches the curve at one point



**Example #3:** Estimate the instantaneous rate of change of  $f(x) = x^2$  at the point  $x = 2$ . We cannot find the slope of the tangent because we only know one point on the tangent line, P.

We can estimate the slope of the tangent at P (i.e. the instantaneous rate of change at P) by finding the slopes of secants for smaller and smaller intervals around P. As the point Q on the function approaches the point P, the slope of the secant approaches the slope of the tangent.



instantaneous rate of change is the slope of tangent

$$\frac{a}{a + 0.001}$$

$$\frac{2}{2 + 0.001}$$

$$x_1$$

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$\rightarrow$  slope of tangent

**Mathematically:**

- (1) select points closer and closer to the point P
- (2) calculate the slope of each secant
- (3) the slopes of the secants will approach a value, which is the slope of the tangent

**Method 1**

Choose a point whose x-value is very close to the one given and calculate the slope using a table of values until a pattern emerges

| Point 1        |                | Point 2        |                | Slope = $\frac{y_2 - y_1}{x_2 - x_1}$ |
|----------------|----------------|----------------|----------------|---------------------------------------|
| x <sub>1</sub> | y <sub>1</sub> | x <sub>2</sub> | y <sub>2</sub> |                                       |
| 2              | 4              | 2.1            | 4.41           | 4.1                                   |
| 2              | 4              | 2.01           |                |                                       |
| 2              | 4              | 2.001          |                |                                       |
| 2              | 4              | 2.0001         |                |                                       |

The slope of tangent is \_\_\_\_\_.

**Method 2**

If we want to know the instantaneous rate of change of  $f(x)$  at  $x = a$ , consider the intervals between  $x = a$  and  $x = a \pm h$ , where  $h$  is a really small number. (We can always consider  $h=0.001$ )

$$\text{Instantaneous Rate of Change} = \frac{f(a+h) - f(a)}{(a+h) - a}$$

$$I.R.O.C = \frac{f(a+h) - f(a)}{h}, \text{ as } h \rightarrow 0$$

or  ~~$f(a) - f(a-h)$~~

$$h = 0.001$$

(This is known as the **difference quotient**.)

$$I.R.O.C = \frac{f(2+0.001) - f(2)}{0.001}$$

= \_\_\_\_\_

**Example #4:** An emergency flare is shot into the air. Its height, in metres, above the ground at various times in its flight is given by the following table.

|            |     |       |      |      |      |      |      |       |      |
|------------|-----|-------|------|------|------|------|------|-------|------|
| Time (s)   | 0.0 | 0.5   | 1.0  | 1.5  | 2.0  | 2.5  | 3.0  | 3.5   | 4.0  |
| Height (m) | 2.0 | 15.75 | 27.0 | 37.1 | 42.0 | 46.8 | 47.0 | 45.75 | 42.0 |

$$a = 2 \quad h = 0.5$$

Estimate the instantaneous rate of change in height at exactly  $t = 2.0$  s.

- Plan: ① Find A.R.O.C between  $[1.5, 2]$   
 ② Find A.R.O.C between  $[2, 2.5]$   
 ③ Find the average of ① and ②

$$\frac{\frac{f(a+h)-f(a)}{a+h-a} + \frac{f(a)-f(a-h)}{a-(a-h)}}{2} = \frac{\frac{f(a+h)-f(a)}{h} + \frac{f(a)-f(a-h)}{h}}{2}$$

$$= \frac{\frac{f(a+h)-f(a-h)}{h}}{2} = \frac{f(a+h)-f(a-h)}{2h}$$

$$a = 2 \quad h = 0.5$$

$$I.R.O.C = \frac{f(2+0.5)-f(2-0.5)}{2(0.5)} = \frac{f(2.5)-f(1.5)}{1} = \frac{46.8-37.1}{1} = 9.7 \text{ m/s}$$

**Example #5:** The height of a soccer ball above the ground at time  $t$  after it is kicked into the air, is given by the formula  $h(t) = -4.9t^2 + 3.5t + 1$  where  $h$  is the height in metres,  $t$  is the time in seconds.

AROC

- (a) Calculate the average velocity of the ball between  $t=0.1$ s and  $t=0.3$ s

$$AROC = \frac{h(0.3) - h(0.1)}{0.3 - 0.1}$$

$$= \frac{1.609 - 1.301}{0.2}$$

$$= 1.54$$

$\therefore$  The ave. velocity is  $1.54 \text{ m/s}$

- (b) Calculate the instantaneous velocity at  $t=0.6$  s.

$$IROC = \frac{f(a+h) - f(a)}{h} \quad ; h = 0.001$$

$$= \frac{f(0.6+0.001) - f(0.6)}{0.001}$$

$$= \frac{f(0.601) - f(0.6)}{0.001}$$

$$\doteq \frac{1.3336151 - 1.336}{0.001}$$

$$\doteq -2.3849$$

$$\doteq -2.4$$

$$h(0.3) = -4.9(0.3)^2 + (3.5)(0.3) + 1$$

$$h(0.3) = 1.609$$

$$h(0.1) = -4.9(0.1)^2 + 3.5(0.1) + 1$$

$$h(0.1) = 1.301$$

$1.54 \text{ m/s}$   
moving upward

$$f(0.601) = 1.3336151$$

$$f(0.6) = 1.336$$

The ball is moving  
down with velocity  
 $2.4 \text{ m/s}$